IMPLEMENTED AND TESTED

Temporal upscaler fixes

The integration defects behind the reported wiggling and detached block-edge lines have been corrected in the tested scenes. The final build also improves fractional transparency history and adds native-resolution hand AA. This addendum records the changes and their practical limits.

Area	Delivered behavior
FSR2 / FSR3 / DLSS	SDK jitter Y now matches raster orientation at each native super-resolution boundary.
DLSS sampling	Phase coverage follows actual input/output pixel area, independently of the TAA slider.
FSR transparency	Fractional history lookup and matching filtered color/opaque samples replace integer snapping.
First-person presentation	Off / 2x / 4x / 8x hand MSAA, cutout coverage and per-sample shading; default 4x.
Packaging	Both mod-owned native bridges rebuilt and verified inside the public Fabric JAR. NVIDIA runtime DLLs remain user supplied.



Five-times nearest-neighbor enlargement of the same isolated block top. Each panel averages 96 frames. The original bright row separated by a dark gap is absent after correction.

Acceptance combines edge fidelity, temporal stability, matched camera paths, GPU input inspection and normal-JAR testing. Low variance alone can reward blur; it was not used as the sole quality criterion.

FSR2 SUPER RESOLUTION

Stable sampling and masks

The FSR2 dispatch now receives the Y sign implied by the actual rendered projection. Frame-generation presentation uses a separate convention and was not globally flipped. [1]

Residual glass variation after that correction led to a second change: previous color and previous opaque color are sampled at the same fractional reprojected coordinate, with linear filtering. The composition and edge tests use the same footprint.

1080p preset	Floor SD before / after	Thin geometry SD	Glass SD
Native AA	0.833 / 0.162	1.871 / 0.758	1.915 / 0.783
Quality	0.304 / 0.188	0.635 / 0.523	0.805 / 0.625
Ultra Performance	0.466 / 0.342	1.152 / 0.978	1.308 / 1.318

Values are mean per-pixel temporal RGB standard deviation on the 0-255 display scale. Frozen fixtures, sharpening 0 and MIP bias Off; fixed-side sequences contain 96 frames. The original audit survey provides the baseline. Nonzero variation on thin or translucent features remains expected.

Additional check	Observed result
Native-AA floor	0.833 to 0.162 SD, about 80.5% lower temporal variation.
Controlled camera translation	The captured motion field reprojects floor surfaces correctly under horizontal, vertical and diagonal motion.
Moving pig body	Using object vectors reduced warp error from 17.09 to 4.22 RGB levels, compared with forcing zero motion.
Particle disappearance	Once the actual input signal fell below 0.1 RGB levels above background, the following output residual was approximately 0.013 to 0.009. No substantial long trail was established.

Fractional lookup improved the measured transparency cases. Additional depth-history resources or a new material-mask capture pass were not added speculatively: the targeted fade and motion tests did not establish a remaining defect requiring those costs. Newly revealed thin geometry and very low input resolutions can still lose detail.

FSR2 Ultra Performance glass SD is 1.318 versus 1.308 in the original survey, about 0.8% higher. It improved from 1.439 in the sign-only test. The repair does not improve every statistic in every preset.

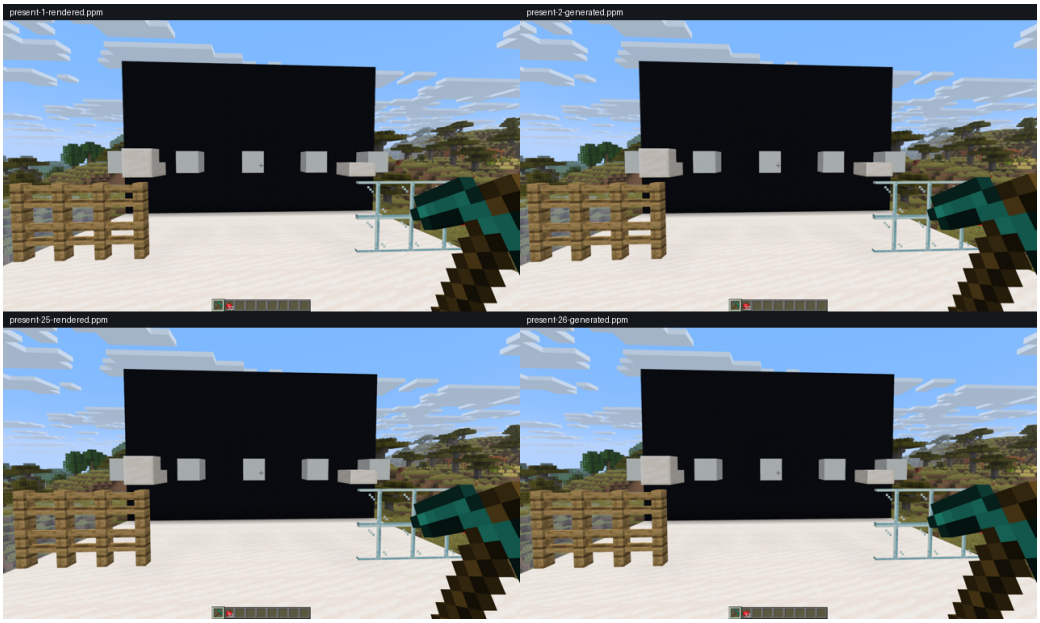
FSR3 SUPER RESOLUTION

Transparency and presentation

FSR3 receives the same super-resolution coordinate correction and fractional mask-history change as FSR2. The separate frame-generation preparation convention is preserved. [2]

1080p preset	Floor SD before / after	Thin geometry SD	Glass SD
Native AA	0.088 / 0.066	0.292 / 0.195	0.700 / 0.282
Quality	0.203 / 0.148	0.435 / 0.311	1.141 / 0.547
Ultra Performance	0.562 / 0.376	1.043 / 0.728	1.722 / 1.283

Fractional history also reduced glass SD relative to the sign-only implementation: 0.350 to 0.282 at Native AA, 0.676 to 0.547 at Quality, and 1.480 to 1.283 at Ultra Performance. This is an incremental comparison after the primary coordinate repair.



Actual presentation readbacks during a camera sweep. The retained window contains 16 rendered and 16 generated frames, all with dispatchReset=false. Images are upright and HUD/hand remain in the intended orientation.

Controlled object motion reduced body warp error from 14.41 to 5.12 RGB levels. Following particle disappearance, measured output residual was about 0.080 to 0.071 RGB levels. FSR3 remained softer than the supersampled reference on some moving fine geometry; that information-loss tradeoff is strongest at Performance and Ultra Performance.

DLSS SUPER RESOLUTION

Correct edges and phase coverage

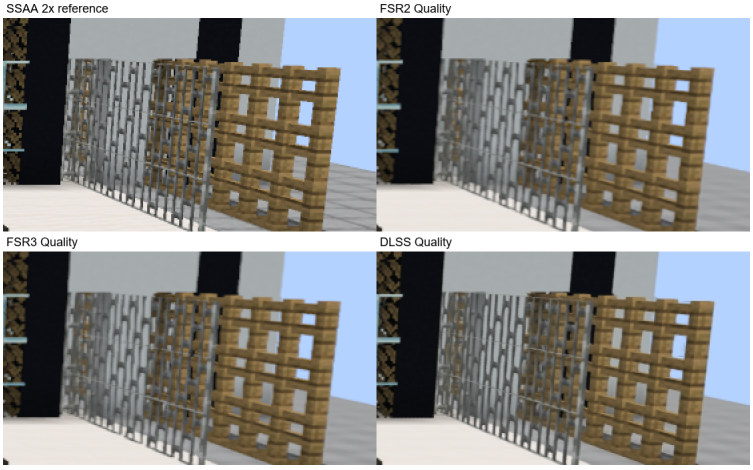
The native Streamline constants now use the corrected raster-to-SDK Y sign. DLSS also gets an automatic Halton cycle of ceil(8 x output pixel area / input pixel area), using the optimal dimensions returned by its runtime. It no longer inherits the capped TAA phase setting. [3,4]

1080p preset	Verified nominal phases
Quality	18
Balanced	24
Performance	32
Ultra Performance	72
Auto	Quality policy at 1080p; actual returned dimensions still determine phases.

Rounding matters: 853 x 480 into 1280 x 720 uses 19 phases with the ceiling rule. Unit tests cover scale, rounding, invalid dimensions and integer-overflow avoidance. Static explicit-preset traces show complete cycles and no repeated reset flag.

1080p preset	Floor SD before / after	Thin geometry SD	Glass SD
Quality	0.079 / 0.057	0.271 / 0.253	0.254 / 0.230
Balanced	0.081 / 0.071	0.275 / 0.310	0.251 / 0.275
Performance	0.290 / 0.156	0.871 / 0.588	0.527 / 0.386
Ultra Performance	0.230 / 0.228	0.444 / 0.477	0.294 / 0.272

On the isolated block, the original top-edge row/gap profile was approximately 56.27 / 32.85 RGB levels. After correction it is approximately 11.67 / 73.66, consistent with a single edge rather than a detached bright line.

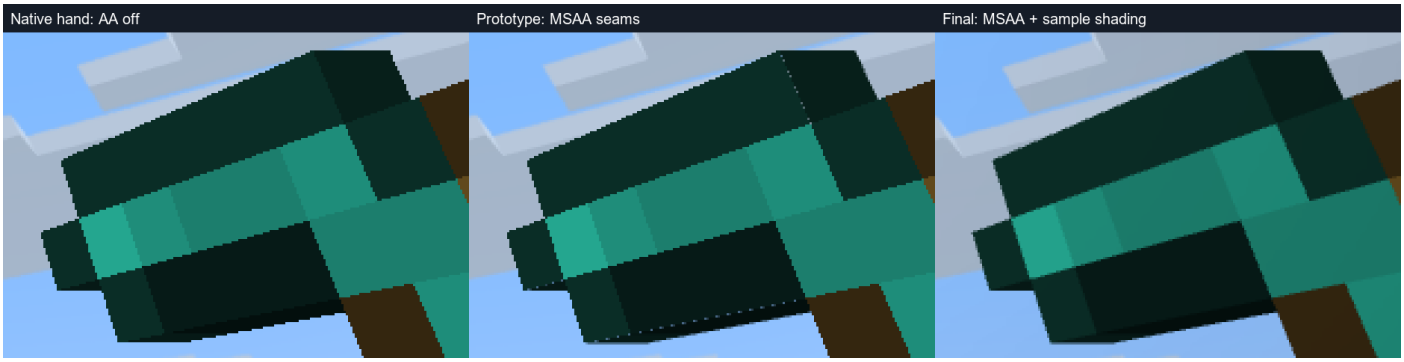


A matched moving-camera frame against 2x SSAA. Reference and upscaler paths have identical captured camera coordinates. This is a common-scene fidelity reference, not a vendor ground-truth or cross-game score.

SHARED FIRST-PERSON RENDERING

Clean hand and item edges

The hand is rendered after world reconstruction at display resolution. Its AA has an independent multisample color/depth target, a pixel-exact world background copy and a color resolve. Hand pixels never enter world temporal history. The HUD stays outside this pass.



Three-times nearest-neighbor crops: vanilla native hand; the rejected pixel-shaded MSAA prototype with dotted seams; and the final per-sample-shaded hand. The final path covers alpha-tested atlas boundaries as well as geometry edges.

Centroid UV interpolation did not remove the prototype seams and was discarded. Per-sample shading fixed them by evaluating texture alpha at the covered samples. The optional Vulkan sampleRateShading feature is requested only when supported; otherwise the hand retains native rendering. Cutout coverage remains separately adjustable. [5]

1080p setting	GPU scopes, summed medians	Hand target memory
Off	No extra hand-AA scopes	0
4x MSAA	0.040 copy + 0.036 render + 0.030 resolve = 0.106 ms	63.3 MiB
8x MSAA	0.062 copy + 0.066 render + 0.037 resolve = 0.165 ms	126.6 MiB

Each enabled-setting measurement used 1,200 frames and 10 sparse timestamp samples per scope on the RTX 3060 Ti. These are pass costs in a simple fixture, not universal overhead or an FPS guarantee. Ordinary hand drawing already has a cost when AA is Off.



Selected frames from a 600-frame live input capture. The hand changes pose without leaving a temporal copy behind. Additional retained checks cover cutout leaves, translucent glass and maps in all three temporal methods.

ACCEPTANCE CHECKS

Normal Fabric JAR validation

Check	Result and scope
Public launch	Fabric production launch, development=false; dependency classpath contains no project source directories. Same public JAR used for tests.
Bridge packaging	Exact packaged DLL bytes match the rebuilt mod bridges. Three native DLLs total: two mod bridges plus AMD runtime. No NVIDIA DLLs or QA harness classes inside the public JAR.
Method and preset coverage	Source-client static survey covers both FSR native-AA modes and explicit upscaling presets; packaged launch covers FSR2, FSR3, DLSS Quality and DLSS Auto. FSR1 and NIS hand rendering also checked.
Resolution changes	Six rapid transitions across 720p, 1080p and 1440p recovered. No device loss or rendering fallback was observed.
Memory lifetime	Hand AA Off and spectator mode release the hand target. After world exit: hasWorld=false, handAaBytes=0, all owned image targets=0.
Missing DLSS DLLs	Cold launch with blank plugin path keeps DLSS selected, disables unavailable controls, shows Set up DLSS and renders normally. Both owned-target counters are 0.
Menus and resources	AA controls inspected at GUI scales 2 and 3; conditional hand coverage row updates immediately. Fullscreen/windowed transition and resource reload returned to valid output.
Special content	Controlled object/camera motion, disocclusions, glass, particles, underwater view, tools, leaves, maps and first-person swing captures reviewed.
Build checks	127 tests; 0 failures; 0 errors. Includes shipped-shader SPIR-V compilation and the public-JAR guard.

The normal run/26.3 configuration and options are byte-for-byte unchanged from the pre-audit backup. All fixtures and diagnostic captures use separate run directories. Test clients have been stopped.

INTERPRETATION AND PROVENANCE

What the results establish

The reproduced mod integration defects are corrected in the retained comparisons. They do not establish zero artifacts in every scene. Temporal upscalers in AAA games also trade stability, detail and motion response; tiny subpixel features, transparency, disocclusions and very low input resolutions remain difficult. Quality is the strongest default comparison point.

The tests use one Windows RTX 3060 Ti and driver 591.86, Minecraft 26.3, Fabric Loader 0.19.5, Fabric API 0.162.0+26.3 and Java 25. AMD/Intel hardware, OpenGL, Sodium/Iris, other resource packs, HDR and all mod combinations were not certified. No Vulkan validation layer was installed. SSAA is a same-scene reference; no literal side-by-side AAA game capture benchmark was performed.

Public artifact

build/libs/salts-anti-aliasing-mc26.3-1.1.jar

SHA-256: 6983ce2a1ec230c48d4d80ae63b87d90e986443388bc50092da7b1b7af03b1e4

Restart Minecraft or Gradle runClient before testing: already loaded native bridge DLLs cannot receive these changes in a running process. Existing user-downloaded DLSS setup can be retained.

Primary implementation references

[1] AMD - FSR2 temporal upscaling integration

[2] AMD - FSR3 super-resolution upscaler

[3] NVIDIA - pinned Streamline 2.14.1 DLSS integration

[4] NVIDIA - DLSS Programming Guide

[5] Khronos - multisampling and fragment shading

Evidence and reproduction

Original audit: output/pdf/salts-upscalers-rendering-audit-2026-10-24.pdf. Fixed-code experiments: build/upscaler-fixes-20261024/. Source-client captures: run/upscaler-fixes-20261024/. Public-JAR captures and per-run settings: run/upscaler-fixes-public-20261025/. The adjacent upscaler-fixes-evidence-2026-10-25/ folder includes enlarged images and validation.json with hashes, measurements and test totals.

Important exclusions: the early SNBT-controlled pig fixture was unreliable on 26.3 and was replaced with direct Java entity creation. Its motion conclusions were discarded. A duplicate final Off performance label was excluded from any repeated-Off inference. The final 4x/8x scope samples remain distinct and valid.